COMSATS University Islamabad

Attock Campus

Department Of Computer Science

Course	***IS(LAB)***
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Assignment No.	***01***

Student Details

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Implement the Caesar Cipher (10 Marks)

Write a Python program that encrypts and decrypts a message using the Caesar Cipher.

1. Write a function caesar_encrypt(text, shift) that takes a string and a shift value as input and returns the encrypted text.

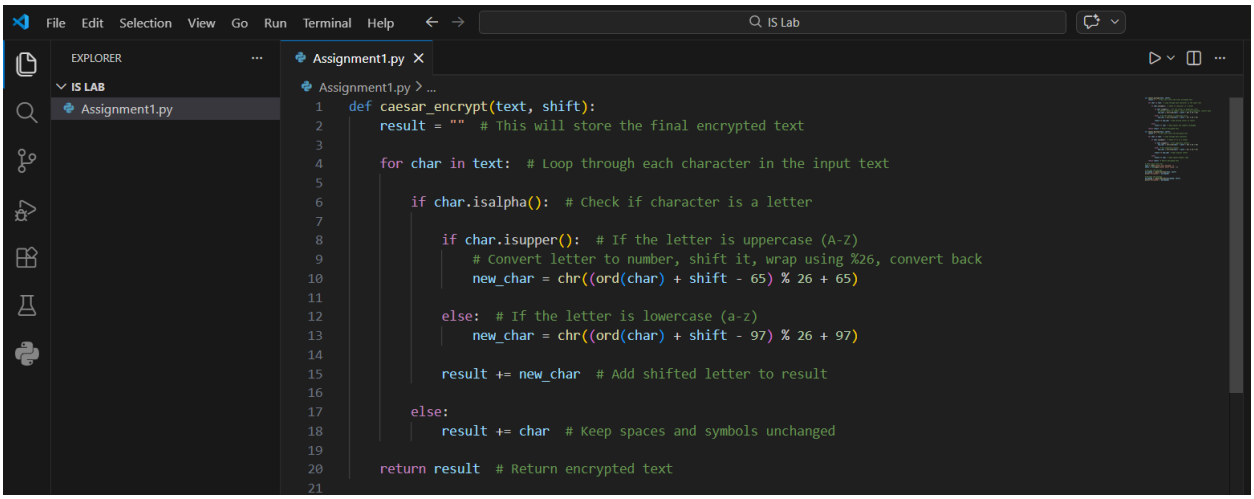
Answer: Encryption Code

```
def caesar_encrypt(text, shift):
    result = ""  # This will store the final encrypted text

    for char in text:  # Loop through each character in the input text

        if char.isalpha():  # Check if character is a letter

            if char.isupper():  # If the letter is uppercase (A-Z)
                # Convert letter to number, shift it, wrap using %26, convert back
                new_char = chr((ord(char) + shift - 65) % 26 + 65)

            else:  # If the letter is lowercase (a-z)
                new_char = chr((ord(char) + shift - 97) % 26 + 97)

            result += new_char  # Add shifted letter to result

        else:
            result += char  # Keep spaces and symbols unchanged

    return result  # Return encrypted text
```

Explanation of Code

The caesar_encrypt(text, shift) function is responsible for converting a normal message into an encrypted form using the Caesar Cipher technique. It starts by initializing an empty string called result, which will store the final encrypted output. The function then iterates through each character of the input text using a loop. For every character, it first checks whether it is an alphabet using char.isalpha(). This ensures that only letters are modified, while spaces, numbers, and special symbols remain unchanged.
If the character is an uppercase letter, the program converts it into its ASCII value using ord(char) and then normalizes it to a range of 0–25 by subtracting 65 (ASCII value of 'A'). The shift value is then added, and modulo 26 is applied to ensure the value wraps around within the alphabet range. After that, 65 is added back to convert it into the correct ASCII range, and chr() is used to transform it back into a character. A similar process is applied to lowercase letters, but using 97 (ASCII value of 'a') instead. The transformed character is then appended to the result string. If the character is not a letter, it is directly added to the result without any modification. Finally, the function returns the fully encrypted string.

Screenshot of Code



2. Write a function caesar_decrypt(ciphertext, shift) that reverses the encryption and returns the original message.

Answer: Decryption Code

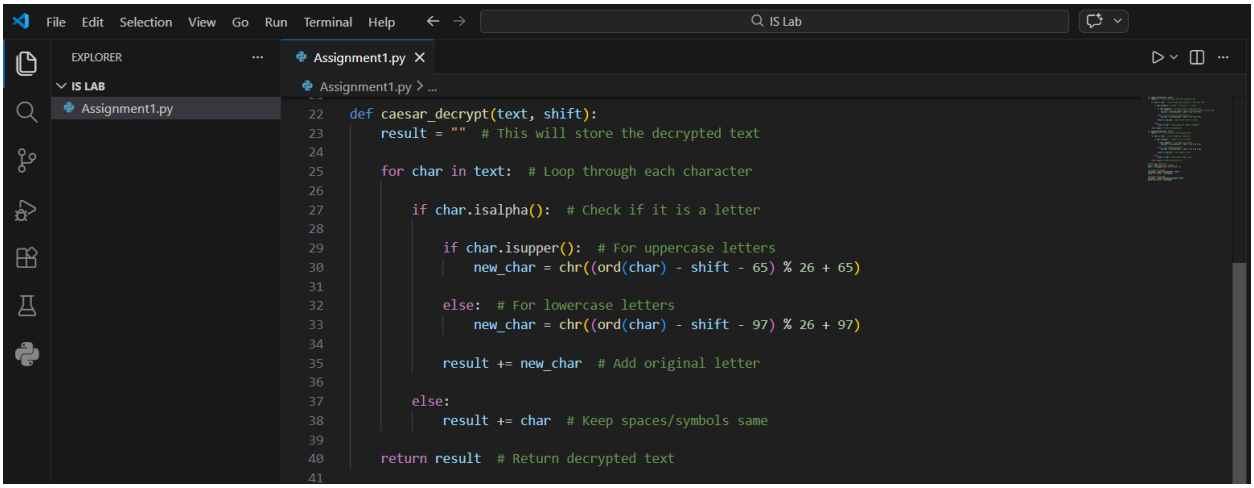
```
def caesar_decrypt(text, shift):
    result = ""  # This will store the decrypted text

    for char in text:  # Loop through each character

        if char.isalpha():  # Check if it is a letter

            if char.isupper():  # For uppercase letters
                new_char = chr((ord(char) - shift - 65) % 26 + 65)

            else:  # For lowercase letters
                new_char = chr((ord(char) - shift - 97) % 26 + 97)

            result += new_char  # Add original letter

        else:
            result += char  # Keep spaces/symbols same

    return result  # Return decrypted text
```

Explanation of Code

The caesar_decrypt(text, shift) function performs the reverse operation of encryption and is used to retrieve the original message from the encrypted text. Like the encryption function, it begins by initializing an empty string called result and iterates through each character of the input ciphertext. It checks whether each character is an alphabet using char.isalpha(), ensuring that only letters are processed while other characters remain unchanged.
For uppercase letters, the function converts the character into its ASCII value and normalizes it to a 0-25 range by subtracting 65. Instead of adding the shift value, it subtracts the shift to reverse the encryption process. The modulo 26 operation is again used to handle wrap-around

cases, ensuring values stay within the alphabet range. After this, 65 is added back and chr() converts the value into the corresponding uppercase letter. The same logic applies to lowercase letters using 97 as the base value. Each decrypted character is appended to the result string. Non-alphabetic characters are added without any changes. In the end, the function returns the decrypted message, which matches the original input text.

Screenshot of Code



3. Take input from the user and print the encrypted/decrypted data.

Answer: Input Code

```
# Take input from user
text = input("Enter your message: ")
shift = int(input("Enter shift value: "))

# Encrypt the message
encrypted = caesar_encrypt(text, shift)
print("Encrypted:", encrypted)

# Decrypt the message
decrypted = caesar_decrypt(encrypted, shift)
print("Decrypted:", decrypted)
```

Explanation of Code

The main program serves as the interface between the user and the encryption/decryption functions. It begins by taking input from the user using the input() function, where the user enters the message they want to encrypt. Then, it asks the user to enter a shift value, which is converted into an integer using int() since numerical operations are required for shifting characters.
After collecting the inputs, the program calls the caesar_encrypt() function, passing the user's message and shift value as arguments. The returned encrypted text is stored in a variable named encrypted and displayed using the print() function. Next, the program calls the caesar_decrypt() function using the encrypted text and the same shift value. This function returns the original message, which is stored in the variable decrypted and printed on the

screen. This demonstrates that the encryption and decryption processes are correctly implemented, as the final output matches the original input.

Screenshot of Code

```python

# Take input from user
text = input("Enter your message: ")
shift = int(input("Enter shift value: "))

# Encrypt the message
encrypted = caesar_encrypt(text, shift)
print("Encrypted:", encrypted)

# Decrypt the message
decrypted = caesar_decrypt(encrypted, shift)
print("Decrypted:", decrypted)
```

Output Of Code

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS

PS D:\BS-SE 04\Information Security\IS Lab> & C:\Users\DELL\AppData\Local\Python\pythoncore-3.14-64\python.exe
"d:/BS-SE 04/Information Security/IS Lab/Assignment1.py"
Enter your message: ali
Enter shift value: 3
Encrypted: dol
Decrypted: ali
PS D:\BS-SE 04\Information Security\IS Lab> & C:\Users\DELL\AppData\Local\Python\pythoncore-3.14-64\python.exe
"d:/BS-SE 04/Information Security/IS Lab/Assignment1.py"
Enter your message: Ahmad Raza
Enter shift value: 3
Encrypted: Dkpdg Udcd
Decrypted: Ahmad Raza
PS D:\BS-SE 04\Information Security\IS Lab>
```